

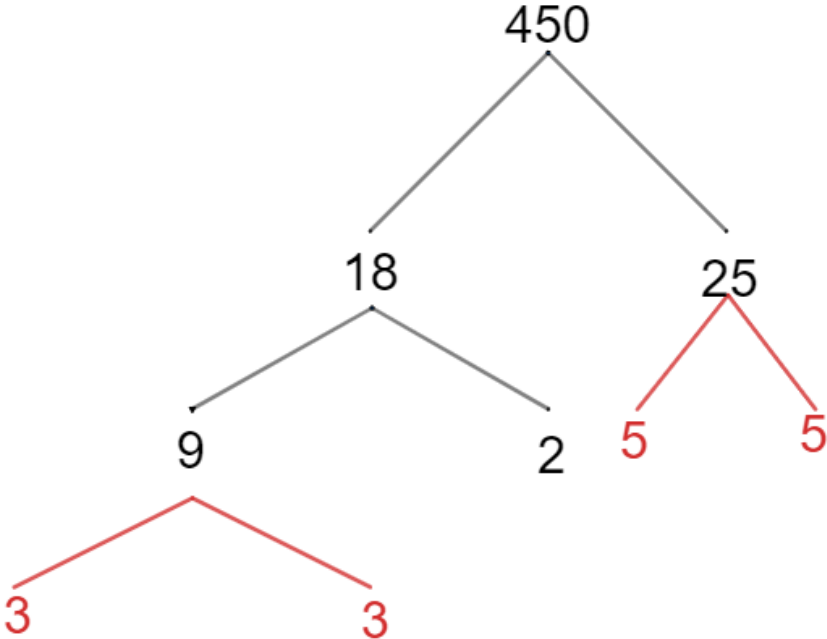
6th Grade Unit 7 Assessment Guide

General Information	The benchmarks on this assessment have assessment limits that may be helpful for the teacher to understand. For example, in the test item specifications, MA.6.NSO.3.3 has an assessment limit that the expression can only have a positive rational base and a natural number exponent. This limit is so that if a student incorrectly answers a question such as $(-3)^2$, it is not because of a student's lack of knowledge of operations with integers (MA.6.NSO.4.1 and MA.6.NSO.4.2). Instruction in unit 7 purposively extended beyond this limit. It is important to remember that assessment limits in Florida's test item specifications are there for statewide assessments and should not be used to limit student learning. Lesson 15 extended past this limit because the benchmark language states integers can be bases and the authors felt that a student could extend their learning of multiplication of integers without a significant jump. Questions 1 and 3 on this assessment match the learning in the unit and match examples for MA.6.NSO.3.3 in the Grade 6 B.E.S.T. Instructional Guide for Mathematics but extend past the assessment limit. Each teacher should consider whether partial credit on questions for the benchmarks MA.6.NSO.4.1 and MA.6.NSO.4.2 are appropriate. If students show that they struggle with operations with integers, then teachers could consider remediation. Math Nation's curriculum continues to build on student learning of operations with integers in units 11 and 12 so a teacher has time for remediation and to build each student's procedural fluency of operations with integers. Teachers will notice that there are no questions that are "a positive plus a positive". This is because this is considered below grade level. The test item specifications for MA.6.NSO.4.1 state that "items involving addition must incorporate at least one negative integer." Questions on this assessment follow this assessment limit.
Calculator Information	All questions are no calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1	Benchmark(s)	MA.6.NSO.3.3
	In the expression $(-7)^5$, the base is -7 and the exponent is 5 , resulting in the repeated multiplication sentence of $-7 \cdot -7 \cdot -7 \cdot -7 \cdot -7$.	Recommended Scoring (5 Total Points)	Suggested points for each part is 2 points for the base, 2 points for the exponent, and 1 point for the multiplication sentence.
		Additional Notes	Students may represent multiplication in various ways. A common error is that students write the multiplication sentence as $-7 \cdot 7 \cdot 7 \cdot 7 \cdot 7$. Consider noting how many of your students make this error and consider how to address this error.

Question	2	Benchmark(s)	MA.6.NSO.3.4
$2^2 \cdot 3^2 \cdot 7$	Recommended Scoring (5 Total Points)		Suggested points is 2 points for 2^2 , 2 points for 3^2 , and 1 point for 7.
	Additional Notes		Consider student work when determining partial credit. If a student makes a minor error, consider following their error to determine if that error is carried through with everything else being correct. In these cases consider deducting just 1 point. Consider awarding 3 points for students who find the correct prime factorization, but do not express it using exponents.

Question	3	Benchmark(s)	MA.6.NSO.3.3
		Recommended Scoring (5 Total Points)	Suggested points are 5 points.
		Additional Notes	Consider deducting 2 points if student has the two correct equivalent expressions but adds one incorrect expression as equivalent, deduct 3 points if student has the two correct equivalent expressions but adds two incorrect expressions as equivalent, and 4 points if student has the three correct equivalent expressions but adds three incorrect expressions as equivalent. Multiselect questions on statewide assessments do not get partial credit.

Question	4	Benchmark(s)	MA.6.NSO.3.4
		Recommended Scoring (5 Total Points)	Suggested points are 5 points.
		Additional Notes	Consider student work for partial credit. Students should be showing knowledge of prime factors on this question. If students show a branch off of 2, the student may not understand that 2 is a prime number.
Question	5	Benchmark(s)	MA.6.NSO.3.3
	(3.71)(3.71)(3.71)(3.71)	Recommended Scoring (5 Total Points)	Suggested points are 5 points.
		Additional Notes	Students may represent multiplication in various ways.

Question	6	Benchmark(s)	MA.6.NSO.4.1
☐ $-12 + (-2)$ ☒ $-12 + 2$ ☐ $-2 + (-12)$ ☐ $2 - (-12)$ ☒ $2 - (12)$ ☐ $12 + (-2)$ ☐ $-2 - (-12)$	Recommended Scoring (5 Total Points)	Suggested points are 5 points.	
		Additional Notes	

This question is assessing procedural fluency of adding and subtracting integers. Consider deducting **2** points if student has the two correct equivalent expressions but adds 1 incorrect expression as equivalent, deduct **3** points if student has the 2 correct equivalent expressions but adds two incorrect expressions as equivalent, and **4** points if student has the 3 correct equivalent expressions but adds three incorrect expressions as equivalent. Multiselect questions on statewide assessments do not get partial credit.

Question	7	Benchmark(s)	MA.6.NSO.3.3
-64		Recommended Scoring (5 Total Points)	Suggested points are 5 points.
		Additional Notes	Consider giving partial credit as follows: • 3 points for a correct value • 2 points for correct sign
Question	8	Benchmark(s)	MA.6.NSO.4.1
7		Recommended Scoring (5 Total Points)	Suggested points are 5 points.

Question	9	Benchmark(s)	MA.6.NSO.4.2
-57		Recommended Scoring (5 Total Points)	Suggested points are 5 points.
Question	10	Benchmark(s)	MA.6.NSO.4.1
232		Recommended Scoring (5 Total Points)	Suggested points are 5 points.
Question	11	Benchmark(s)	MA.6.NSO.4.2
9		Recommended Scoring (5 Total Points)	Suggested points are 5 points.
Question	12	Benchmark(s)	MA.6.NSO.4.2
96		Recommended Scoring (5 Total Points)	Suggested points are 5 points.

Question	13	Benchmark(s)	MA.6.NSO.4.1
$-5,071$		Recommended Scoring (5 Total Points)	Suggested points are 5 points.
Question	14	Benchmark(s)	MA.6.NSO.4.1
52		Recommended Scoring (5 Total Points)	Suggested points are 5 points.
Question	15	Benchmark(s)	MA.6.NSO.3.3
$\frac{9}{49}$		Recommended Scoring (5 Total Points)	Suggested points are 5 points.

Question	16	Benchmark(s)	MA.6.NSO.4.2
-6		Recommended Scoring (5 Total Points)	Suggested points are 5 points.
Question	17	Benchmark(s)	MA.6.NSO.4.1
3		Recommended Scoring (5 Total Points)	Suggested points are 5 points.
Question	18	Benchmark(s)	MA.6.NSO.4.1
0		Recommended Scoring (5 Total Points)	Suggested points are 5 points.
		Additional Notes	Student understanding of additive inverses was introduced as zero pairs. If a number of students miss this question, then consider revisiting zero pairs.

Question	19	Benchmark(s)	MA.6.NSO.3.3
0.0016		Recommended Scoring (5 Total Points)	Suggested points are 5 points.
		Additional Notes	Students multiply a multi-digit number with decimals to the tenths by one-tenth and one-hundredth with procedural reliability in grade 5. Units 2 and 3 continue a student's learning of multiplication of decimals. If students are showing issues with place value, then consider remediation.
Question	20	Benchmark(s)	MA.6.NSO.4.2
−225		Recommended Scoring (5 Total Points)	Suggested points are 5 points.